PROJECT DOCUMENTATION – STEP-BY-STEP PROJECT DEVELOPMENT PROCEDURE

PROJECT TITLE

VOICE BASED CONCEPT UNDERSTANDING ANALYSER

Step 1: Problem Identification

In educational environments, evaluating a student's conceptual understanding through spoken explanations is often a manual and time-consuming process. Teachers must listen to each student's explanation, assess the correctness of concepts, evaluate communication skills, and provide feedback.

Traditional assessment methods are subjective and require significant effort. There is a need for an AI-powered system that can automatically analyze spoken explanations, measure conceptual understanding, and provide instant feedback.

This project addresses the challenge by developing a Voice Based Concept Understanding Analyser that combines Speech-to-Text, Natural Language Processing (NLP), Semantic Similarity Analysis, and Speech Fluency Evaluation to automatically assess users' conceptual understanding.

Step 2: Problem Statement

To develop an AI-powered web application that evaluates a user's conceptual understanding from spoken explanations by converting speech into text, comparing it with a predefined reference answer, calculating semantic similarity, analyzing speech quality, and generating structured feedback.

Step 3: Objectives

The objectives of the project are:

- Develop an AI-powered voice-based assessment system.
- Convert speech into text using OpenAI Whisper.
- Analyze semantic similarity between spoken explanations and reference concepts.
- Evaluate conceptual understanding using Sentence-BERT embeddings.
- Calculate similarity scores using Cosine Similarity.
- Assess speech fluency and communication quality.

- Generate qualitative feedback based on understanding level.
- Display transcription and evaluation results through a Streamlit interface.
- Generate downloadable PDF reports.
- Deploy the application for educational use.

Step 4: Data Collection

The project uses two primary sources of data:

Voice Input

- User-recorded audio (.wav/.mp3)

Reference Concepts

Predefined concept explanations such as:

- Machine Learning
- Cloud Computing
- Artificial Intelligence
- Data Science
- Deep Learning

Dataset Summary

- Audio Files: User Uploaded
- Reference Concepts: Text Format
- Missing Values: None

Step 5: Environment Setup
The application was developed using Python.
Development Environment:

- Google Colab
- Visual Studio Code

Required Libraries

- Streamlit
- OpenAI Whisper

- Sentence Transformers
- Scikit-learn
- NumPy
- ReportLab
- Pydub

Step 6: Data Preparation

Before evaluation, the following preprocessing steps were performed:

- Audio upload
- Audio format validation
- Speech transcription
- Text cleaning
- Lowercase conversion
- Removal of unwanted symbols
- Sentence formatting

The processed text becomes the input for semantic similarity analysis.

Step 7: Speech-to-Text Conversion

The uploaded audio is converted into text using the OpenAI Whisper model.

Procedure

1. Upload voice recording.
2. Load Whisper model.
3. Transcribe speech into text.
4. Display generated transcription.

This module provides accurate speech recognition for educational explanations.

Step 8: Semantic Understanding Analysis

The transcribed explanation is compared with the predefined reference answer.

Procedure

1. Load Sentence-BERT model.
2. Generate embeddings for reference text.
3. Generate embeddings for spoken explanation.
4. Compute Cosine Similarity.
5. Calculate Concept Understanding Score.

This score reflects how closely the user's explanation matches the expected concept.

Step 9: Speech Fluency Evaluation

The application also evaluates speech quality.

Evaluation Parameters

- Speech clarity
- Completeness
- Fluency
- Semantic consistency
- Overall communication quality

Based on these parameters, users receive performance feedback.

Step 10: Feedback Generation

The system generates automated feedback.

Example Feedback

Excellent Understanding

- Similarity Score ≥ 80%

- Good Understanding
- Similarity Score 60–79%

Average Understanding

- Similarity Score 40–59%

Needs Improvement

- Similarity Score < 40%

Feedback also includes recommendations for improvement.

Step 11: Streamlit User Interface

A user-friendly interface was developed using Streamlit.

Features

- Audio Upload
- Speech-to-Text Display
- Concept Similarity Score
- Understanding Level
- Feedback Panel
- Download Report Button

The interface provides a simple workflow for students and educators.

Step 12: PDF Report Generation

The application generates a downloadable PDF report containing:

- User Transcription
- Reference Concept
- Similarity Score

- Understanding Level
- Speech Feedback
- Final Evaluation

This report can be used for educational assessment.

Step 13: Testing

The application was tested for:

- Audio upload functionality
- Speech recognition accuracy
- Semantic similarity calculation
- Streamlit interface responsiveness
- PDF report generation
- Overall application performance

Step 14: Results

The Voice Based Concept Understanding Analyser successfully:

- Converted speech into text accurately.
- Evaluated conceptual understanding using AI.
- Generated semantic similarity scores.
- Provided instant qualitative feedback.
- Generated downloadable PDF reports.
- Improved assessment efficiency for educators.

Step 15: Conclusion

The Voice Based Concept Understanding Analyser d

emonstrates the effectiveness of Artificial Intelligence in educational assessment. By combining Speech Recognition, Natural Language Processing, Semantic Similarity Analysis, and Interactive Visualization, the system provides an automated, objective, and scalable method for evaluating conceptual understanding.
The project reduces manual evaluation effort while improving consistency and providing meaningful feedback to learners.

Step 16: Future Scope
Future enhancements may include:

- Real-time voice recording
- Multi-language speech recognition
- AI-generated personalized learning suggestions
- Integration with Learning Management Systems (LMS)
- Student performance analytics dashboard
- Cloud deployment using Google Cloud Platform
- Mobile application support
- Emotion and confidence detection during speech
- Voice biometrics for secure student authentication

Technologies Used

- Python
- Streamlit
- OpenAI Whisper
- Sentence Transformers (Sentence-BERT)
- Scikit-learn
- NumPy
- ReportLab
- Google Colab
- Visual Studio Code

This documentation is specifically tailored for your **Voice Based Concept Understanding Analyser** SmartBridge project and follows the same professional format as the sample document you uploaded.